B1.1 Carbohydrates and lipids

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b1-1.html

Written for this site from the class slides. The chapter is on pages 181–204 of the course textbook.
Image credits are on the last slide of the slides for this subtopic.

Carbon and polymers

B1.1.1 Why carbon is the basis of life

pp. 181–204

A covalent bond is a strong bond formed by sharing electrons. A carbon atom can form up to four single bonds, or a mix of single and double bonds, with carbon or other non-metals. So it can form unbranched and branched chains, and single or multiple rings.

B1.1.2 Condensation

pp. 181–204

Condensation links monomers into a polymer, releasing a water molecule for each link. Examples are polysaccharides, polypeptides and nucleic acids.

B1.1.3 Hydrolysis

pp. 181–204

Hydrolysis digests polymers into monomers. Water is split, and it supplies the -H and -OH groups. This happens in digestion, and in the surroundings of decomposers.

Carbohydrates

B1.1.4 Monosaccharides

pp. 181–204

In ring form, **pentoses** have 5 carbons, and **hexoses** have 6. Glucose is **soluble** and **transportable**, it is **chemically stable**, and it **yields energy** when oxidized in respiration.

B1.1.5 Starch and glycogen

pp. 181–204

Starch in plants and glycogen in animals are **compact**, because they coil and branch. They are relatively **insoluble**, because of their size, so they have little osmotic effect. Alpha-glucose monomers are easily added or removed, to build or mobilize a store.

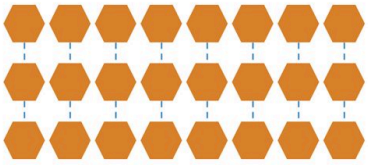
B1.1.6 Cellulose

pp. 181–204

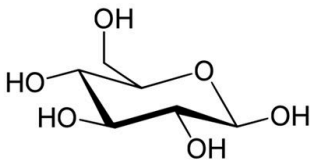
Beta-glucose monomers alternate in orientation, which gives **straight chains**. Chains are grouped in bundles and cross-linked by **hydrogen bonds**, which gives high strength for plant cell walls.

Cellulose: a structural polysaccharide

B1.1.6



straight beta-glucose chains linked by hydrogen bonds (blue)



Beta-glucose. Yikrazuul, public domain.

Cellulose is made of **beta-glucose**: each unit is flipped relative to the next, so the chain is **straight**. Straight chains lie side by side, joined by **hydrogen bonds** into strong **microfibrils** in cell walls.

Key idea
Alpha or beta glucose changes the shape of the polymer, and so its job.

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B1.1.7 Glycoproteins

pp. 181–204

A **glycoprotein** is a protein with a carbohydrate chain, which faces out from the membrane. Other cells can recognize it. **ABO antigens** on red blood cells differ by the sugars at the ends of their chains.

Lipids

B1.1.8 Lipids are hydrophobic

pp. 181–204

Lipids dissolve in non-polar solvents, and are only sparingly soluble in water. They include fats, oils, waxes and steroids.

B1.1.9 Forming triglycerides and phospholipids

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By condensation, one glycerol links to **three fatty acids** in a triglyceride, or to **two fatty acids and a phosphate group** in a phospholipid.

B1.1.10 Saturated and unsaturated fatty acids

pp. 181–204

Type	C=C bonds	Melting point
Saturated	None	Highest: chains pack closely
Mono-unsaturated	One	Lower: a kink in the chain
Poly-unsaturated	Two or more	Lowest: more kinks

Plants store energy in oils, and endotherms store it in fats, which suits their warm bodies.

B1.1.11 Triglycerides in adipose tissue

pp. 181–204

Triglycerides suit **long-term energy storage**: they are insoluble and store about twice as much energy per gram as carbohydrate, though they mobilize slowly. They are also **thermal insulators**, especially as blubber in endotherms in cold water.

B1.1.12 Phospholipid bilayers

pp. 181–204

A phospholipid is **amphipathic**: it has a hydrophilic phosphate head and two hydrophobic tails. In water, heads face out and tails face in, so a **bilayer** forms.

B1.1.13 Steroids pass through the bilayer

pp. 181–204

Steroids, such as **oestradiol** and **testosterone**, have four fused rings. They are non-polar, so they can pass through the hydrophobic core of a membrane.

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